

Droplet Size Evolution in Clouds

Microphysics Simulation using PySDM

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Project Objectives



- Simulate the lifecycle of cloud droplets from **nucleation** to **rainfall**.
- Analyze the evolution of the **Droplet Size Distribution (DSD)**.
- Study the efficiency of the **Super-Droplet Method (SDM)**.
- Implementation using the PySDM Python library.

Core Microphysical Processes



1. Nucleation

Activation of aerosol particles into small water droplets.

2. Condensation

Growth of droplets due to water vapor diffusion in supersaturated air.

3. Collision-Coalescence

Stochastic process where droplets collide and merge into larger rain drops.

Simulation Goals



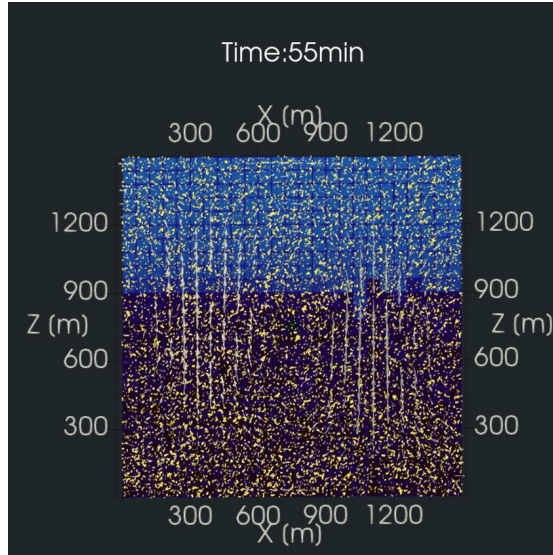
- ① **DSD Broadening:** Observe how the droplet spectrum widens over time.
- ② **Precipitation Timing:** Identify the "Time-to-Rain" (formation of 0.2mm drops).
- ③ **Aerosol Impact:** Compare clean vs. polluted air scenarios.

Simulation Setup & Experiments



- **Core Simulation:** Configured a 2D cloud microphysics model using the PySDM framework to simulate droplet evolution from nucleation to warm rain.
- **Physical Sensitivity Runs:** Conducted comprehensive experiments to analyze how environmental data affects droplet growth:
 - *Aerosols & Chemistry:* CCN concentration (Twomey effect) and hygroscopicity (κ).
 - *Thermodynamics:* Moisture profiles, potential temperature, and vertical updraft velocity.
- **Numerical Testing:** Performed stability checks across different grid resolutions, time steps (dt), and super-droplet densities (n_{sd}).

Simulation Video Output



Exp 1 & 2: Aerosol Loading and Water Vapor Content

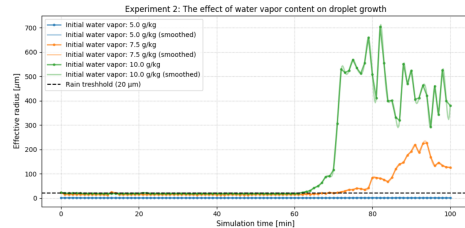
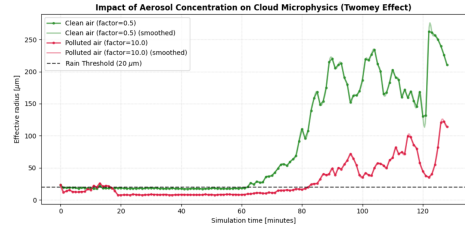


Aerosols (Twomey Effect)

Clean air drives rapid, massive rainout ($265 \mu\text{m}$). Extreme pollution locks water into a stable regime, heavily delaying precipitation.

Initial Water Vapor

Low moisture (5.0 g/kg) starves cloud formation completely. High moisture (10.0 g/kg) triggers torrential, runaway rainout ($700 \mu\text{m}$).



Exp 3 & 4: Vertical Convection and Aerosol Chemistry

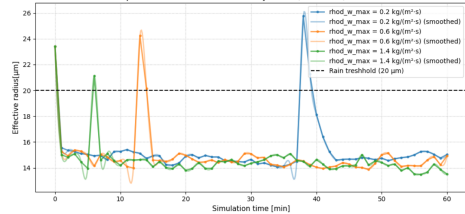
Convective Dynamics

Strong updrafts act as kinetic accelerators. Rapid adiabatic cooling triggers the earliest localized rain cascades (at 6 min vs 38 min).

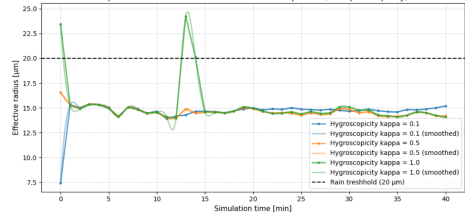
Aerosol Chemistry (κ)

High hygroscopicity ($\kappa = 1.0$) drastically lowers activation barriers for early growth. Low hygroscopicity ($\kappa = 0.1$) limits the total number of activated droplets, leaving more water mass available per droplet to reach a higher baseline.

Experiment 3: The effect of vertical dynamics (convection) on clouds



Experiment 4: The effect of aerosol chemical composition (Absorption capacity)



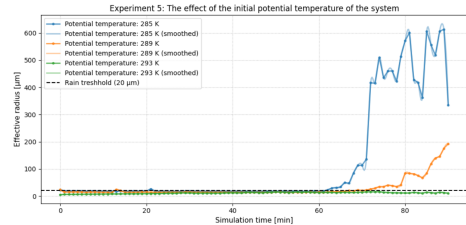
Exp 5: Initial Potential Temperature



Thermodynamic Trade-offs

Governed by the Clausius-Clapeyron relation.

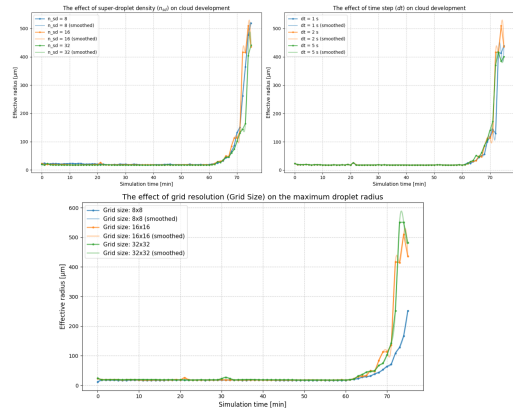
- **Cool Air (285 K):** Lowers the saturation barrier, enabling massive supersaturation that drives explosive droplet growth ($620 \mu\text{m}$).
- **Warm Air (293 K):** Retains moisture in the vapor phase, rendering the cloud colloidally stable and entirely rainless.



Mathematical Testing Results



- Super-Droplet Density (n_{sd}):**
 $n_{sd} = 32$ smooths out Monte Carlo noise and stabilizes the statistical tail against artificial growth spikes.
- Time Step (dt):** $dt = 1$ s provides the necessary temporal fidelity to handle stiff equations without truncation errors.
- Grid Size:** A refined 32×32 mesh eliminates numerical diffusion, properly resolving updrafts and preventing unphysical droplet trapping.



Comprehensive Conclusion



Key Takeaway

Macroscopic cloud properties (precipitation timing, intensity) are critically sensitive to micro-scale environmental and thermodynamic inputs.

The Physics-Numerics Bond

In advanced particle-based simulations like PySDM, physics and numerics are deeply intertwined. Deploying highly accurate physics without tight spatial (**32x32**) and temporal (**$\Delta t = 1$ s**) model convergence inevitably triggers severe numerical artifacts (e.g., artificial droplet trapping).

Outcome

Achieving a precise balance between high-fidelity microphysical parameters and converged grid kinematics is the only viable pathway to predictive cloud modeling.

 PySDM Documentation
<https://pysdm.github.io/>

 Shima et al. (2009)
The super-droplet method for the numerical simulation of clouds and precipitation.
QUARTERLY JOURNAL OF THE ROYAL METEOROLOGICAL SOCIETY